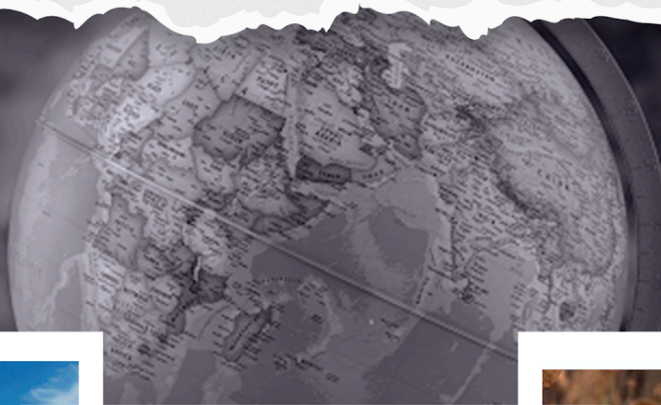


General Knowledge

World Geography

STUDY NOTES

Geomorphology: Introduction



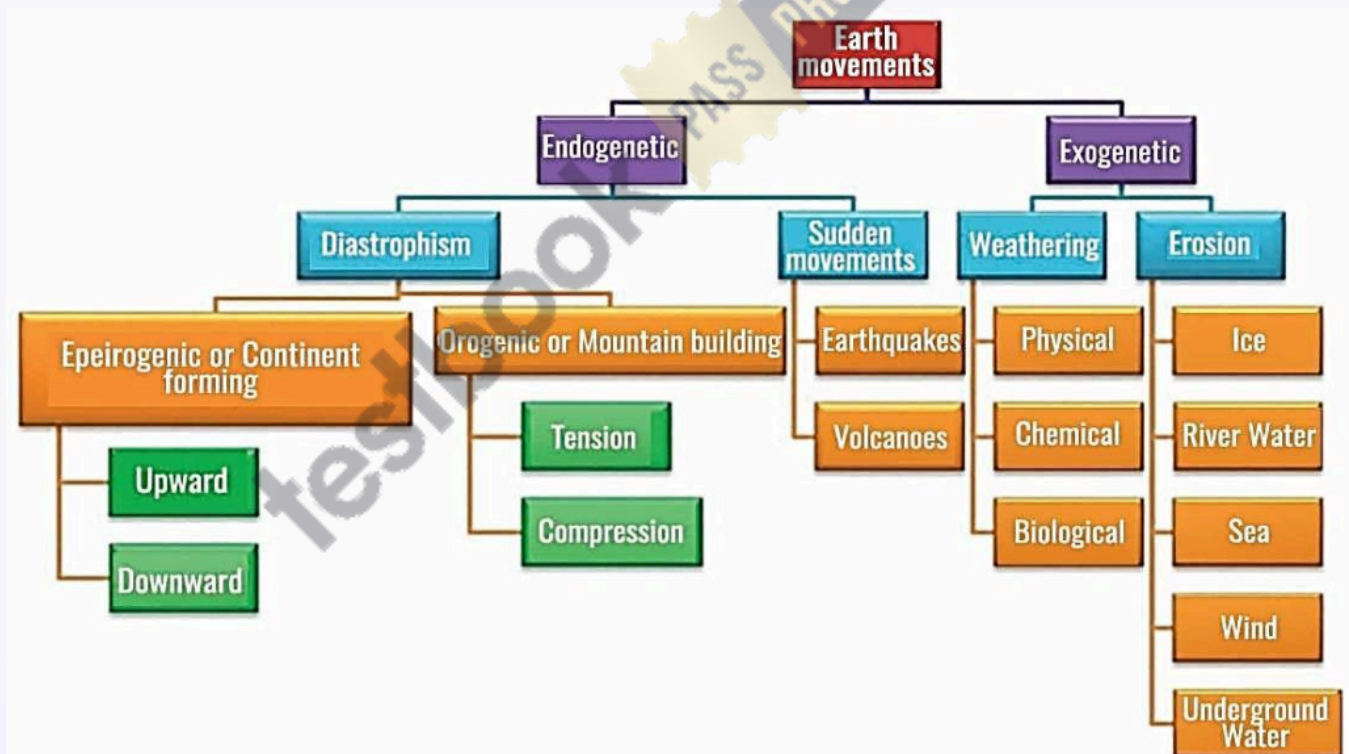
Geomorphology: Introduction

Geomorphology is the scientific study of landforms and the processes that shape them. **William Morris Davis (1850 - 1934)** was a U.S. geographer, geologist, and meteorologist who founded the science of geomorphology.

The surface of the Earth is uneven due to the dynamic nature of the Earth's crust. The crust is continually shaped by internal (endogenic) and external (exogenic) forces.

- **Endogenic Forces (Internal Forces):** These forces, such as tectonic movements and geothermal activity, originate from within the Earth and result in the elevation and building up of the Earth's surface.
- **Exogenic Forces (External Forces):** These forces, driven by energy from the sun (solar radiation), continuously wear down the surface through processes like erosion and weathering.

The continuous interaction between these internal and external forces keeps the Earth's surface uneven. Endogenic forces (land-building) elevate parts of the Earth's surface, while exogenic forces (land-wearing) erode or degrade elevations. This ongoing interaction results in the persistence of surface variations over time.



Geomorphic Processes

Geomorphic processes are the physical and chemical actions that shape the Earth's surface. These processes are divided into **endogenic** and **exogenic** types:

1. **Endogenic Processes:** These involve forces from within the Earth, such as:
 - **Diastrophism:** Movements within the Earth's crust, including orogeny (mountain building) and epeirogeny (continental uplifting).

- **Volcanism:** The eruption of molten rock and gases from beneath the Earth's crust, contributing to surface build-up.
- 2. **Exogenic Processes:** These are surface processes that wear down the Earth's surface and include:
 - **Weathering:** The breakdown of rocks into smaller particles due to physical or chemical processes.
 - **Mass Wasting:** The downslope movement of rock, soil, and debris under the influence of gravity.
 - **Erosion:** The removal of surface materials (rock, soil) by agents like wind, water, ice, and biological activity.
 - **Deposition:** The accumulation of eroded materials in lower areas, such as basins or valleys.

Geomorphic Agents

Geomorphic agents are natural forces or elements that cause erosion, transport materials, and deposit them in new locations. These agents include:

- **Water:** Rivers, streams, and rainfall.
- **Ice:** Glaciers.
- **Wind:** Air currents that carry sand and dust.
- **Waves and Currents:** Movement of water along coastlines.

These agents are crucial in shaping the Earth's surface through the processes of erosion, transportation, and deposition.

The Role of Gravity

Gravity plays a critical role in all geomorphic processes. It provides the directional force that activates the movements of materials:

- **Downslope Movements:** Gravity causes materials to move from higher to lower areas.
- **Stress Activation:** Gravity contributes to the stresses that lead to ocean currents, winds, and other surface movements.

Without gravity, no material movement or surface processes like erosion would occur.

Endogenic Processes: Diastrophism and Volcanism

- **Diastrophism:** Refers to the movement, elevation, or deformation of the Earth's crust. It includes:
 - **Orogeny:** Mountain-building processes through severe folding of the crust, affecting narrow belts of the Earth's surface.
 - **Epeirogeny:** Large-scale uplift or warping of broad sections of the Earth's crust, leading to the formation of continents.
 - **Earthquakes:** Sudden, minor movements that cause localized shifts in the crust.
 - **Plate Tectonics:** Horizontal movement of tectonic plates that can cause shifts in the Earth's surface.
- **Volcanism:** The eruption of magma, ash, and gases from the Earth's interior, contributing to the construction of landforms like volcanoes and lava plateaus.

Differences between Epeirogeny and Orogeny

- **Orogeny:** Involves intense crustal deformation, leading to the formation of mountain ranges. This process is more localized and occurs along narrow belts.
- **Epeirogeny:** A more gradual and widespread uplifting of large continental areas, causing broad deformations without the severe folding seen in orogeny.

Impact of Endogenic and Exogenic Forces

Endogenic forces build and elevate the Earth's surface, while exogenic forces wear it down, creating a balance that results in an uneven surface. Over long periods, these processes have shaped the Earth's topography and continue to influence it today. Understanding these processes is crucial for managing the Earth's resources sustainably and preserving its potential for future generations.

Endogenic Geomorphic Movements

- “Endo” means internal; “genic” means origin; “geo” means earth and “morphic” means form.
- The large-scale movements of the earth's crust or its surface brought down by the forces emanating from deep below the earth's surface are **called endogenic geomorphic movements or simply endogenic movements.**

The force behind Endogenic Movements

- The ultimate source of energy behind forces that drive endogenic movements is **earth's internal heat.**
- Earth's internal heat is a result of mainly radioactive decay which is a source for 50% of the earth's internal heat and gravitational pressure gradients.
- Differences in temperature gradients or geothermal gradients and pressure gradients among various layers of the earth give rise to density differences and these density differences give rise to **conventional currents.**
- Convection currents in the mantle drive the crust and upper mantle and the movement of the lithospheric plates is the cause behind endogenic movements.
- The **Coriolis effect** can influence where convection currents travel.
- The destination of convection currents determines the nature and location of the endogenic movements.

Classification of Endogenic Movements

- Endogenic movements are divided into **diastrophic movements** and **sudden movements.**
- Diastrophism refers to **deformation** of the Earth's crust. Diastrophic movements are gradual and might stretch for thousands of years.
- On the other hand, sudden movements like earthquakes and volcanic eruptions occur in a very short period.

- Diastrophic movements are further classified into **epeirogenic movements** (continent forming — subsidence, upliftment) and **orogenic movements** (mountain building — folding, faulting).

Diastrophism

- Diastrophism refers to deformation of the Earth's crust due to movements such as **folding, faulting, warping and fracturing**.
- All processes that move, elevate or build up portions of the earth's crust come under diastrophism. They include –
 1. **Epeirogenic movements**
 - Epeirogenic or **continent-forming** movements are **radial** movements that act **along the radius of the earth**.
 2. **Orogenic or mountain-forming movements**
 - In contrast to epeirogenic movement, the orogenic movement is a **more complicated deformation** of the Earth's crust, associated with convergence of tectonic plates.
 - Such plate convergence forms orogenic belts that are characterised by "the folding and faulting of layers of rock, by the intrusion of magma, and by volcanism.
 - Orogenic or the mountain-forming movements **act tangentially to the earth surface**, as in plate tectonics.



Sudden Movements

- Sudden geomorphic movements occur mostly at the **tectonic plate margins**.
- The plate margins are highly unstable regions due to pressure created by pushing and pulling of magma in the mantle's **convection currents**.
- These movements cause considerable deformation over a short period.
- They include –
 - 1) **Earthquakes**

- o An earthquake in Chile (1822) caused a one-metre uplift in coastal areas.
- o An earthquake in New Zealand (1885) caused an uplift of up to 3 metres.

2) Volcanoes

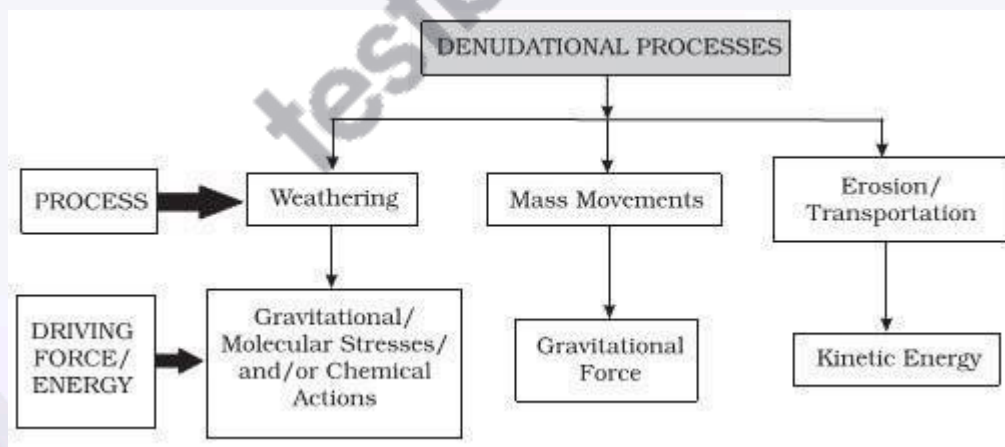
- Depending on the chemical composition and viscosity of the lava, a volcano may take various forms.

Exogenic Geomorphic Movements

- The geomorphic processes on the earth's crust or its surface brought down by the forces emanating from **above the earth's surface** like wind, water etc. are called exogenic geomorphic processes.
- Exogenic geomorphic processes give rise to exogenic geomorphic movements or simply exogenic movements such as **weathering** and **erosion**.
- The effects of most of the exogenic geomorphic processes are small and slow but will, in the long run, affect the rocks severely due to continued fatigue.

The force behind Exogenic Movements

- Exogenic processes are a direct result of the sun's heat. The sun's energy dictates the weather patterns like winds, precipitation, etc.
- Sun's heat along with weather patterns are responsible for stress-induced in earth materials giving rise to exogenic movements mainly **weathering** and **erosion**.
- All the exogenic processes i.e. weathering and erosion are covered under a general term, **denudation**. The word 'denude' means to strip off or to uncover.



Weathering -

Weathering is the **disintegration** of rocks, soil, and minerals under the influence of heat, pressure and chemical processes like leaching, oxidation and reduction, and hydration.

- There are three major groups of weathering processes: **1) chemical, 2) physical or mechanical and 3) biological weathering processes.**

Physical Weathering Processes

- Physical weathering involves **mechanical disintegration** of rocks due to temperature changes, freeze-thaw cycles, wet-dry cycles, crystallisation of salts, animal and plant activity, etc.

Various mechanisms of physical weathering are explained below:

1) Exfoliation due to pressure release or unloading

- **Intrusive rocks formed deep beneath the Earth's surface are under tremendous pressure due to overlying load and removal of the overlying load** because of continued erosion causes vertical pressure release with the result that the upper layers of the rock expand and fracture parallel to the surface.
- Over time, sheets of rock break away from the exposed rocks along the fractures, a process known as **exfoliation**.

2) Exfoliation due to thermal stress weathering

- **Thermal stress weathering results from the subsequent expansion and contraction of rocks** caused by diurnal and seasonal variations in temperatures.
- The surface layers of the rocks tend to expand more than the rock at depth, and this leads to peeling off of the surface layers.
- This process is most effective in **dry climates and high elevations** where diurnal temperature changes are drastic.

3) Granular Disintegration

- Granular disintegration happens in rocks composed of different types of coarse-grained minerals.
- **Dark-coloured minerals absorb more heat than the light-coloured minerals.** This leads to differential expansion and contraction of mineral grains resulting in grain-by-grain separation from the rock.

4) Frost weathering

- **Frost weathering occurs due to the growth of ice within pores and cracks of rocks** during repeated cycles of freezing and melting.
- Frost weathering is the collective name for several processes **where ice is present.**
- These processes include **frost shattering, frost-wedging and freeze-thaw weathering.**

5) Salt Weathering

- Salt weathering occurs when saline solutions seep into cracks and joints in the rocks and evaporate, leaving salt crystals behind.
- Salt crystals expand during the crystallization process and also when they are subjected to above-normal temperatures.
- The expansion in near-surface pores causes splitting of individual grains within rocks, which eventually fall off (granular disintegration or granular foliation).
- Salt weathering is normally associated with **arid climates** where strong heating causes strong evaporation and crystallisation.

6) Mass Wasting

- Mass wasting is the mass movement of unconsolidated soil, sand, rocks, etc. along a slope under the influence of gravity.
- Mass wasting occurs when the gravitational force acting on a slope exceeds its resisting force leading to **slope failure**.

Chemical Weathering

- Chemical weathering involves the **chemical decomposition** of rocks and soil.
- Chemical weathering processes include **dissolution, solution, carbonation, hydration, oxidation and reduction** which act on the rocks to decompose, dissolve or reduce them to a fine state.
- These weathering processes are interrelated and go hand in hand and hasten the weathering process.

1) Natural dissolution

- Dissolution is a process where a solute in a gaseous, liquid, or solid phase dissolves in a solvent to form a solution.
- Some minerals, due to their natural **solubility** like nitrates, sulphates, and potassium; and **oxidation potential** will weather through dissolution naturally during rains.
- These minerals are easily **leached** out without leaving any residue and accumulate in dry regions.

2) Solution weathering

- Solution weathering occurs when the solvent is an **acidic solution** rather than simple water. **Acidic solutions** are any solution that has a higher concentration of hydrogen ions than water; solutions that have a lower concentration of hydrogen ions than water is called **basic or alkaline solutions**.

3) Carbonation

- Carbonation weathering is a process in which atmospheric carbon dioxide leads to **solution weathering**.

- As rain falls, it dissolves small amounts of carbon dioxide from the air, forming a weak acid that can dissolve some minerals like **limestone (calcium carbonate)**.
- When carbonic acid reacts with limestone, it produces calcium bicarbonate, which is partially soluble in water (dissolution weathering).
- Caves are formed when underground water containing carbonic acid travels through blocks of limestone, dissolves out the limestone, and leaves empty pockets (caves) behind (E.g. **Karst topography**).

4) Hydration

- Hydration is the **chemical addition of water** that involves the rigid attachment of H^+ and OH^- ions to the atoms and molecules of a mineral.
- When rock minerals take up water, the increased volume creates physical stresses within the rock. For example, iron oxides are converted to iron hydroxides which are larger in volume.

5) Hydrolysis

- In biological hydrolysis, a **water molecule is consumed** to affect the **separation of a larger molecule into component parts**.
- In biological hydrolysis, pure water reacts with **silicate** or **carbonate minerals** resulting in the complete dissolution of the original mineral (dissolution weathering).

Biological activity and weathering

- Biological weathering is the removal of minerals from the environment due to growth or movement of organisms. Living organisms contribute to both mechanical and chemical weathering.
- **Lichens** and **mosses** grow on essentially bare rock surfaces and create a more humid chemical microenvironment.
- On a larger scale, seedlings sprouting in a crevice and plant roots exert physical pressure as well as providing a pathway for water and chemical infiltration.
- Burrowing and wedging by organisms like earthworms, rodents etc., help in exposing the new surfaces to chemical attack and assist in the penetration of moisture and air.
- Decaying plant and animal matter helps in the production of **humic, carbonic and other acids** which enhance decay and solubility of some elements.
- Algae utilises mineral nutrients for growth and helps in the concentration of iron and manganese oxides.

Significance of weathering

- Weathering is the **first step in the formation of soil** from rocks.
- Weathering weakens soil and rocks and makes it easy to exploit natural resources.
- Weathering leads to **natural soil enrichment**.

- Weathering leads to **mineral enrichment** of certain ores by leaching unwanted minerals leaving behind the valuable ones.

Erosion

Soil erosion is the loosening and displacement of topsoil from the land due to the action of agents like wind and water. Soil erosion in nature may be a slow process or a fast process promoted by human activities like overgrazing, and deforestation. Weathering and erosion lead to the simultaneous process of 'degradation' and 'aggradation'.

Water Erosion

- Running water is one of the main agents, which carries away soil particles. Soil erosion by water occurs by means of raindrops, waves or ice.
- Erosion by water is termed differently according to the intensity and nature of erosion: **raindrop erosion, sheet erosion, rill and gully erosion, stream bank erosion, landslides, coastal erosion, glacial erosion.**

1) Raindrop erosion or splash erosion

- Raindrops behave like tiny bombs when falling on exposed soil, displace soil particles and destroy soil structure.
- Presence of vegetation on land prevents raindrops from falling directly on the soil thus erosion of soil in areas covered by vegetation is prevented.

2) Sheet erosion

- With continued rainfall the displaced soil particles fill in the spaces between soil particles and prevent water from seeping into the soil. This results in surface runoff and even more erosion.
- The detachment and transportation of soil particles by flowing rainwater is called sheet or wash off erosion.

3) Rill and gully erosion

- In **rill** erosion finger-like rills appear on the cultivated land after it has undergone sheet erosion. These rills are usually smoothened out every year while forming.
- **Gully** erosion is the removal of soil along drainage lines by surface water runoff.
- When rills increase in size, they become gullies. Once started, gullies will continue to move by headward erosion or by slumping of the side walls.

4) Streambank erosion

- The erosion of soil from the banks (shores) of the streams or rivers due to the flowing water is called bank erosion.
- In certain areas where the river changes its course, the river banks get eroded at a rapid rate.
- Streambank erosion damages the adjoining agricultural lands, highways and bridges.

5) Coastal erosion

- In the coastal areas, waves dash along the coast and cause heavy damage to the soil. During the landfall of cyclones, storm surges destroy beaches and wash away the top layer.
- In estuaries, tidal bores cause extensive damage to the surrounding banks.

6) Glacial erosion

- In the polar regions and high mountainous regions like the Himalayas, soil erosion is caused by moving glaciers. This is called glacial erosion.

Wind Erosion

- Wind erosion or **Aeolian erosion** is quite significant in arid and semi-arid regions. Winds usually blow at high speeds in deserts due to the absence of physical obstruction. These winds remove the fertile, arable, loose soils leaving behind a depression devoid of topsoil.
- The depression formation in deserts is the first step in Oasis formation. Oasis forms in depressions when there is underground water that accumulates above rocks.
- Very fine and medium sands are moved by the wind in a succession of bounds and leaps, known as **saltation**.
- Small sand and dust particles are transported over long distances through the air by a process known as **suspension**.
- Coarse sand is not usually airborne but rather is rolled along the soil surface. This type of erosion is called **surface creep**.
- Very coarse sand and gravel are too large to be rolled by wind, so wind-eroded soils have surfaces covered with coarse fragments. This kind of arid soil surface is known as desert pavement.